

Supplementary Methods

Whole Genome and Structural Analysis of *Acinetobacter baumannii* SRR5891469

S1. Dataset Acquisition

Whole genome sequencing data for *Acinetobacter baumannii* isolate SRR5891469 were obtained from the NCBI Sequence Read Archive (SRA).

Organism: *Acinetobacter baumannii*

SRA Accession: SRR5891469

Platform: Illumina HiSeq 2500

Reference Genome: NZ_CP045110.1

S2. Quality Assessment

FastQC v0.12.1 was used for read quality assessment.

Command: fastqc SRR5891469.fastq.gz

S3. Read Trimming

FastP v0.24.0 was used to remove low-quality bases and adapters.

S4. Reference-Based Alignment

BWA v0.7.17 was used for alignment against NZ_CP045110.1.

S5. Alignment Processing

SAMtools v1.21 was used for BAM conversion, sorting and indexing.

S6. Variant Calling

BCFtools v1.21 was used for SNP and variant detection.

S7. Genome Annotation

Prokka v1.14.6 was used for genome annotation.

S8. Antimicrobial Resistance Gene Detection

ABRicate was used with CARD database. Genes detected included OXA-1117, ADC-158, ANT(3'')-IIa, sul2, AbaF, AbaQ, AmvA, AdeABC, AdeFGH and AdelJK.

S9. Virulence Factor Analysis

VFDB analysis identified 110 virulence-associated genes.

S10. OXA-23 Mutation Analysis

Annotated OXA-23 gene (NJBHLHLK_01820; coordinates 1887843–1888667) contained substitutions S64G, A96P, D134H, N146Y, Y153D, P170T, L171I, G240E, S246A and K252N.

S11. Mutation Visualization

IGV v2.x was used for visualization of mutations at NZ_CP045110.1:1887843–1888667.

S12. Protein Structure Prediction

SWISS-MODEL was used for homology modeling of OXA-23.

S13. Structural Validation

PROCHECK: 92.1% favored regions; ERRAT: 94.61 quality factor; Verify3D: 88.32% acceptable residues.

S14. Workflow Summary

NCBI SRA Dataset → FastQC → FastP → BWA-MEM → SAMtools → BCFtools → Prokka → ABRicate (CARD/VFDB) → OXA-23 Extraction → Mutation Analysis → IGV Validation → SWISS-MODEL → PROCHECK + ERRAT + Verify3D → Integrated Analysis